

Paper Manager pack — OpenVLA-OFT (Oct 26)

Paper: Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success

arXiv: <https://arxiv.org/abs/2502.19645>

Site/code: <https://openvla-oft.github.io> · github.com/moojink/openvla-oft

PDF: /workspace/papers/openvla-oft-2502.19645.pdf

Session: Oct 26 · due 9:30 AM · **Stakeholder=3** (only filled); full five packs

Sources: Summarizer + Senior Research (/workspace/openvla-oft-quiet/DELIVERABLE.md)

1. One-liner (Stakeholder)

OFT = speed+success FT patch making OpenVLA viable at 25–50 Hz (26–43× throughput, LIBERO 97.1%) — modernize OpenVLA ops, **don’t** substitute for $\pi_{0.5}$ on BEHAVIOR.

OFT = FT recipe, not a new foundation model. Not FAST. Not FM.

2. Output signature shift (CLEAR)

	Vanilla OpenVLA	OFT
Decode	Serial AR discrete bins	**Parallel** continuous decode
Chunk	K=1	K= **8** LIBERO / K= **25** ALOHA
Head	256-bin CE	Continuous MLP + **L1**
Hz	~4.2	~ **109.7** LIBERO (26×); ALOHA ~ **77.9** (43×)
LIBERO SR	76.5%	**97.1%** (π_0 94.2%)

OFT+ adds FiLM on SigLIP+DINOv2 (multi-view language) — without FiLM, language → ~chance.

3. Exact OFT recipe

1. Parallel decoding (bidirectional attn over empty action embeds)

- 9/6/26, 7:50 AMOpenVLA-OFT — Stakeholder Pack
2. Action chunking K=8/25

3. Continuous MLP head

4. L1 on [-1,1]

5. LoRA r=32; optional wrist+proprio

6. OFT+: FiLM on both vision towers
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4. Role packs

Stakeholder (18+3) — WEIGHT 3 · LONGEST

Board table: LIBERO SR 76.5 → 97.1; Hz 4.2 → 109.7; ALOHA Hz 1.8 → 77.9 (π_0 still 291); scoop 100 vs π_0 93.3; pot 51 vs 42.

Rule of thumb

- OpenVLA/HF/LoRA ops + high Hz → **OFT mandatory**; stock AR obsolete for 25–50 Hz
- Max wall-clock + OpenPI/JAX → $\pi_0/\pi_{0.5}$ still win latency + pretrain coverage
- Tokenizer-only AR fix → **FAST** (different lever)

Choose OFT when: Prismatic/HF codebase; simple L1 FT; LIBERO-style + moderate demos; 7B AR-lineage ablations.

Choose $\pi_{0.5}$ when: OpenPI standardized; long-horizon open-world; **2026 BEHAVIOR primary**.

Risks: fine-tuning 2024 OXE AR $\neq$ 2025 FM generalist; L1 unimodality; FiLM tax; ALOHA chunk latency 0.321s vs π_0 0.086s; “97% LIBERO” $\neq$ BEHAVIOR-ready; FAST $\neq$ OFT confusion.

Scientific Reviewer (8+3)

Strengths: factorial decode $\times$ rep $\times$ objective; throughput+latency; FiLM ablation; L1 $\approx$ diffusion cheaper.
Caveats: LoRA OFT vs full-FT π_0 ; latency abstract “ms” vs tables in seconds; pot ~50%; L1 multimodal undertested. Verdict: better OpenVLA FT default — does **not** license “OFT supersedes FM.”

Empiricist (11+3)

Priority	Experiment
P0 Y	Vanilla vs OFT Cont-L1 dry infer + signature dump

P1 Partial	LIBERO-Spatial LoRA; PD&AC discrete vs Cont-L1; $K \in \{1,4,8\}$; wrist+proprio; vs π_0 /DP/ACT
P2	FiLM; FAST AR vs OFT parallel
Defer	Paper 8×80GB full suites; real ALOHA

Budget: P0 <1 GPU-h; LIBERO scout 16–40 GPU-h. Env: /workspace/opencvla-oft-quiet/env_outline/.

Archaeologist (11+3)

OpenVLA AR bins → **OFT** continuous parallel chunks || **FAST** (discrete AR upgrade) || $\pi_0/\pi_{0.5}$ (FM).
ACT supplies chunk+L1 ancestry. Parallel improvement paths — do not collapse names.

Visionary (11+3)

OFT = corrective hygiene for OpenVLA stacks. Primary BEHAVIOR substrate remains $\pi_{0.5}$ /**OpenPI**.
Keep OFT in harness as open AR-family control. Skip Discord/Drive.

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	**18+3**	**Decision tree OFT vs $\pi_{0.5}$ (HIGH)**
Reviewer	8+3	Not superseding FM
Empiricist	11+3	Signature dump + K sweep
Archaeologist	11+3	OFT FAST FM forks
Visionary	11+3	Harness vs BEHAVIOR deploy

Spine: output signature shift; OFT ≠ FAST ≠ FM. PDF for Mac.

GitHub title: [ECE 605] - OpenVLA-OFT Making Stakeholder

Paths

Artifact	Path
This pack	`/workspace/opencvla-oft-quiet/KANTA_PACK.md`
Senior Research	`/workspace/opencvla-oft-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/opencvla-oft-2502.19645-summary.md`
Paper PDF	`/workspace/papers/opencvla-oft-2502.19645.pdf`